

✓ Shark Tank (continued)

In the first part of this lab, you started to explore the Shark Tank data set (<https://dlsun.github.io/pods/data/sharktank.csv>).

In this second part, you will go through the steps of data analysis and produce a short report. The questions in this part are deliberately more open-ended because we want you to get practice with:

- identifying relevant data questions
- telling a story with the data.

Your answers may differ from your friend's, and that's okay!

Ignore this cell and do not run it manually!

- This cell is necessary to get plotly plots to show when you print the notebook.
- This cell should only be run when you are getting ready to submit your notebook.

Show code

✓ Help Your Friends!

Your friend(s) are determined to appear on *Shark Tank*. While they currently lack a company, a business model, or even a vague concept, they are confident that data can fix this. As an up-and-coming data scientist, you have been asked to help them maximize their odds of getting a deal.

Read in the data into a Pandas `DataFrame` and clean the data like you did in the previous part of this lab (this should only require copy and pasting).

```
# Copy your code here
```

✓ Question 1. How Does Funding Rate Depend on Industry and Gender?

Your friend tells you that they are trying to find a co-founder as well as the industry in which to focus their company.

Use the grammar of graphics to make a **single** visualization that explores how funding rate depends on the industry of the company and the gender of the founders.

Consider design choices that will make your visualization easier to read.

[**We are expecting:** Your code, the legible visualization, and a short paragraph summarizing the conclusions from your visualization.]

```
# Code for your visualization here.
```

YOUR SUMMARY HERE

✓ Question 2. Your Turn to Brainstorm

Considering the data that you have at your disposal, what are a few relevant questions that you believe could help your friend on *Shark Tank*. Come up with at least three questions that can be answered with the data and are relevant to your friend's problems that haven't already been answered in this lab.

[**We are expecting:** Your three questions and what in the data can be used to answer them]

YOUR QUESTIONS HERE

- 1.
- 2.

2.

3.

✓ Question 3. Your Turn!

Answer at least two of your own questions from Question 2 using the data. At least one should be plotted.

[**We are expecting:** Your code, at least one visualization, and a couple sentences describing the takeaways from your data exploration.]

```
# Your code here.
```

Answers to your Questions

✓ Question 4. Present

From any part of the data analysis you have done in Lab 2, choose the most important takeaway that you'd like to communicate to your friend.

[**We are expecting:** One sentence about your chosen takeaway.]

YOUR TAKEAWAY HERE

Create a new data visualization that clearly argues the point you want to share with your friend.

Consider how best to display your takeaway. If it was previously visualized, consider changes you could make to enhance your argument (are there elements you can either add or remove to highlight the takeaway?). If it wasn't, how could you plot it to cleanly display the takeaway?

[**We are expecting:** One of the most beautiful data visualizations we have ever seen, ready to be shared with the

world. It should be clear from the visualization what the takeaway is without having any outside context.]

```
# Code for your visualization here.
```

Write one paragraph explaining your design choices in making your visualization. Why do these choices support the story you're trying to tell? (How did you highlight the takeaway?)

[**We are expecting:** a paragraph describing your design choices and why you made them.]

YOUR EXPLANATION HERE.

✓ Submission Instructions

- **If your final visualization did not render, please take a screenshot and add it to your submission PDF.**
- Restart this notebook and run the cells from beginning to end.
 - Go to Runtime > Restart and Run All.

Run this cell to download this notebook as a webpage, `_NOTEBOOK.html`.

Show code

- Open `_NOTEBOOK.html` in your browser, and save it as a PDF.
 - Go to File > Print > Save as PDF.
- Double check that all of your code and output is visible in the saved PDF.
- Upload the PDF to [Gradescope](#).
 - Please be sure to select the correct pages corresponding to each question